EDA ON AUSTO AUTOMOBILE CUSTOMER DATA

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1 PROJECT CONTEXT

Austo Motor Company, a prominent automaker offering a range of SUVs, Sedans, and Hatchbacks, has recently begun reviewing the performance of its marketing initiatives. During a board discussion, leadership expressed concerns about whether the current approach is achieving its intended impact. To gain clearer insights and strengthen future campaigns, the company has decided to bring in an analytics expert to conduct a detailed evaluation.

2 OBJECTIVES

The objective of this analysis is to study the customer data and understand overall demand patterns. The company aims to identify trends that can help improve customer experience and support better business decisions. The data is examined to answer key questions provided by the Data Science team, and the insights gained will guide improvements in marketing and operations.

3 DATA DESCRIPTION

- Age:The age of the individual in years.
- Gender: The gender of the individual, categorized as male or female.
- Profession: The occupation or profession of the individual.
- Marital_status: The marital status of the individual, such as married &, single
- Education: The educational qualification of the individual Graduate and Post Graduate
- No_of_Dependents: The number of dependents (e.g., children, elderly parents) that the individual supports financially.
- Personal_loan: A binary variable indicating whether the individual has taken a personal loan "Yes" or "No"
- House_loan: A binary variable indicating whether the individual has taken a housing loan "Yes" or "No"
- Partner_working: A binary variable indicating whether the individual's partner is employed "Yes" or "No"
- Salary: The individual's salary or income.
- Partner_salary: The salary or income of the individual's partner, if applicable.
- Total_salary: The total combined salary of the individual and their partner (if applicable).
- Price: The price of a product or service.
- Make: The type of automobile

4 DATA OVERVIEW

4.1 IMPORT THE LIBRARIES

Essential Python libraries are imported to support data handling, analysis, and visualization. These libraries provide built-in functions that make it easier to process and explore the dataset. This forms the basis for the entire analysis.

4.2 LOAD THE DATA

The dataset is loaded into a DataFrame so that all records can be viewed and analysed in a structured format. This step allows the data to be accessed for cleaning, exploration, and further processing.

Here Customer_Data is the DataFrame name and I read the data file named 'austo_automobile.csv' from the Google Drive using the path.

```
      Age  Gender Profession Marital_status      Education  No_of_Dependents  \
0      53    Male   Business        Married  Post Graduate                 4
1      53   Femal   Salaried        Married  Post Graduate                 4
2      53  Female   Salaried        Married  Post Graduate                 3
3      53  Female   Salaried        Married       Graduate                 2
4      53    Male   Salaried        Married  Post Graduate                 3
...   ...     ...        ...            ...            ...               ...
1576   22    Male   Salaried         Single       Graduate                 2
1577   22    Male   Business        Married       Graduate                 4
1578   22    Male   Business         Single       Graduate                 2
1579   22    Male   Business        Married       Graduate                 3
1580   22    Male   Salaried        Married       Graduate                 4

     Personal_loan House_loan Partner_working  Salary  Partner_salary  \
0               No         No             Yes   99300         70700.0
1              Yes         No             Yes   95500         70300.0
2               No         No             Yes   97300         60700.0
3              Yes         No             Yes   72500         70300.0
4               No         No             Yes   79700         60200.0
...            ...        ...             ...     ...             ...
1576            No        Yes              No   33300             0.0
1577            No         No              No   32000             NaN
1578            No        Yes              No   32900             0.0
1579           Yes        Yes              No   32200             NaN
1580            No         No              No   31600             0.0

      Total_salary  Price       Make
0           170000  61000        SUV
1           165800  61000        SUV
2           158000  57000        SUV
3           142800  61000        SUV
```

FIG 1: LOAD THE DATA OUTPUT

4.3 CHECK THE STRUCTURE OF THE DATA

The structure is examined to understand the number of rows, columns, and the general layout of the dataset. This helps in evaluating the size of the data and identifying the different features available for analysis.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1581 entries, 0 to 1580
Data columns (total 14 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Age                 1581 non-null   int64
 1   Gender              1528 non-null   object
 2   Profession          1581 non-null   object
 3   Marital_status      1581 non-null   object
 4   Education           1581 non-null   object
 5   No_of_Dependents    1581 non-null   int64
 6   Personal_loan       1581 non-null   object
 7   House_loan          1581 non-null   object
 8   Partner_working     1581 non-null   object
 9   Salary              1581 non-null   int64
 10  Partner_salary      1475 non-null   float64
 11  Total_salary        1581 non-null   int64
 12  Price               1581 non-null   int64
 13  Make                1581 non-null   object
dtypes: float64(1), int64(5), object(8)
memory usage: 173.1+ KB
```

FIG 2: CHECK THE STRUCTURE OF THE DATA

The dataset contains 1581 rows and 14 columns. Most variables are complete, but some columns contain missing values that require attention.

4.4 CHECK THE TYPES OF THE DATA

Each column's data type is reviewed to understand whether it is numerical, categorical, or textual. Knowing the type helps determine which statistical methods or visualizations are appropriate for analysis.

	0
Age	int64
Gender	object
Profession	object
Marital_status	object
Education	object
No_of_Dependents	int64
Personal_loan	object
House_loan	object
Partner_working	object
Salary	int64
Partner_salary	float64
Total_salary	int64
Price	int64
Make	object

FIG 3: CHECK THE TYPES OF THE DATA

The column types include:

- Numeric (int64/float64): Age, No_of_Dependents, Salary, Partner_salary, Total_salary, Price.
- Categorical (object): Gender, Profession, Marital_status, Education, Personal_loan, House_loan, Partner_working, Make.

4.5 CHECK FOR AND TREAT MISSING VALUES

The dataset is checked to identify any missing entries that may affect analysis. If missing values are found, suitable methods such as imputation or removal are applied to maintain data accuracy and consistency.

	e
Age	0
Gender	53
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	106
Total_salary	0
Price	0
Make	0

FIG 4: CHECK FOR MISSING VALUES OUTPUT

- Missing values in Gender (53 missing values) may affect gender-based insights and should be imputed or categorized carefully. Mode imputation (most frequent category) is commonly used for categorical variables
- Missing values in Partner_salary(106 missing values) align with non-working partners and can be safely replaced based on business logic.This is expected because not all customers have a working partner.

If Partner_working = No, partner salary should naturally be 0.

If Partner_working = Yes, missing values need further investigation or imputation using median salary of partners

Gender	count
Male	1199
Female	327
Femal	1
Femle	1

dtype: int64

FIG 5: TREATING GENDER COLUMN

We identified inconsistent spellings in the Gender column (‘Femal’, ‘Femle’) and corrected them by replacing all variations with the standard label ‘Female’ to ensure data consistency.

The missing values in the Gender column were imputed using the mode (most frequent category), ensuring no loss of data and maintaining consistency in the dataset.

```
#To find Partner salary with zero value
(Customer_Data['Partner_salary']==0).sum()
```

```
np.int64(623)
```

FIG 6: TREATING PARTNER COLUMN

Partner_salary missing values = 106

Total rows in dataset = 1581

Zero values in Partner_salary = 623

Therefore there are 623 zeros in Partner Salary + 106 NaN values,which means that a total of 729 rows (623 + 106) have unusable partner salary information. This is almost 46% of the dataset,so drop method we cannot use since that will harm the dataset structure.

Partner_salary can be recalculated by using the below formula

Partner_salary=Total_salary-Salary

4.6 CHECK THE STATISTICAL SUMMARY

A summary of the numerical columns is reviewed to understand the central tendency, spread, and distribution of values. This helps in detecting potential outliers and understanding the general behaviour of the data.

	count	mean	std	min	25%	50%	75%	max
Age	1581.0	31.922	8.426	22.0	25.0	29.0	38.0	54.0
No_of_Dependents	1581.0	2.458	0.943	0.0	2.0	2.0	3.0	4.0
Salary	1581.0	60392.220	14674.825	30000.0	51900.0	59500.0	71800.0	99300.0
Partner_salary	1581.0	19233.776	19670.391	0.0	0.0	25100.0	38100.0	80500.0
Total_salary	1581.0	79625.996	25545.858	30000.0	60500.0	78000.0	95900.0	171000.0
Price	1581.0	35597.723	13633.637	18000.0	25000.0	31000.0	47000.0	70000.0

FIG 7: CHECK THE STATISTICAL SUMMARY

To enhance readability, the round () function was applied to restrict the decimal precision in the descriptive summary.

4.7 CHECK FOR AND TREAT DATA IRREGULARITIES

The dataset is examined for inconsistencies such as incorrect spellings, outliers, or unusual patterns. If required, these irregularities are corrected or handled to ensure reliable analysis.

To check the data irregularities like outliers or negative values, boxplot is used.

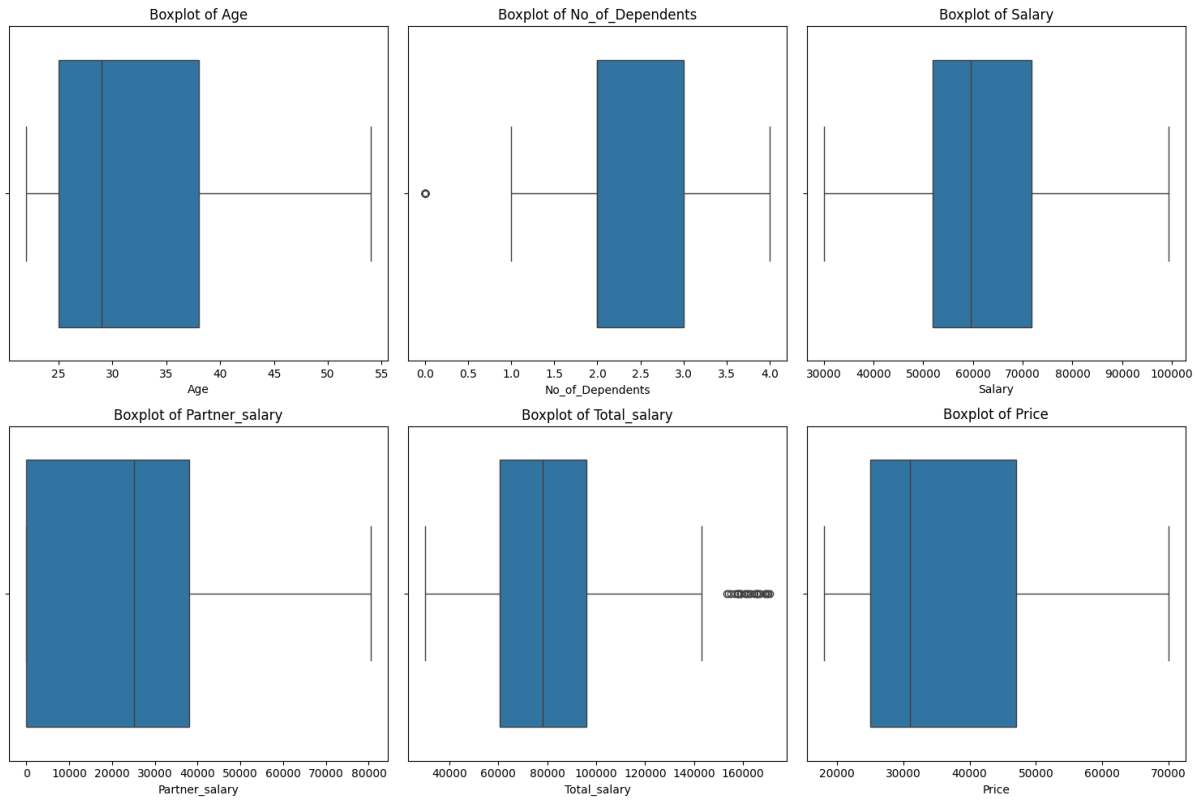

FIG 8: CHECK FOR AND TREAT DATA IRREGULARITIES

Boxplot of Total_salary typically shows:

- Right-skewness
- Large variability
- High-earning outliers

So there is no need to remove these outliers because: They represent important customer segments.High-income families may prefer SUVs or premium models.They help marketing understand premium buyers.Removing them would distort insights.There is no indication that these values are incorrect

4.8 OBSERVATIONS AND INSIGHTS

- **The dataset contains 1581 rows and 14 columns**, covering demographic information, income details, and automobile purchase attributes. The mix of numerical and categorical variables makes the dataset suitable for customer demand analysis.
- **Missing values were found mainly in the Gender and Partner_salary columns.** Gender had a few missing entries, which were filled with the most common category. Partner_salary had more missing values, mostly due to non-working partners, and was treated using appropriate logic.
- **Data types are correctly assigned**, with numerical fields stored as integers or floats and categorical information represented as object types. This ensures proper application of statistical and visualization techniques.
- **Data irregularities such as spelling mistakes in the Gender column** (e.g., “Femal”, “Femle”) were detected and corrected to maintain consistency. Cleaning these issues prevents errors during grouping and analysis.
- **Statistical summaries show a wide spread in Salary, Partner_salary, Total_salary, and Price.** Several variables display right-skewness, and high-income outliers were retained because they represent important premium customer segments rather than data errors.
- **Overall data quality is good**, with only minor corrections needed. After handling missing values and irregularities, the dataset is clean, consistent, and ready for detailed univariate and bivariate analysis.

5 UNIVARIATE ANALYSIS

Each variable in the dataset is examined individually to understand its distribution and characteristics. Histograms, bar plots, and frequency counts are used to study how values are spread. This helps identify common patterns, dominant categories, and initial trends within the data. Histogram is used here to explore.

5.1 EXPLORE ALL THE VARIABLES (CATEGORICAL AND NUMERICAL) IN THE DATA

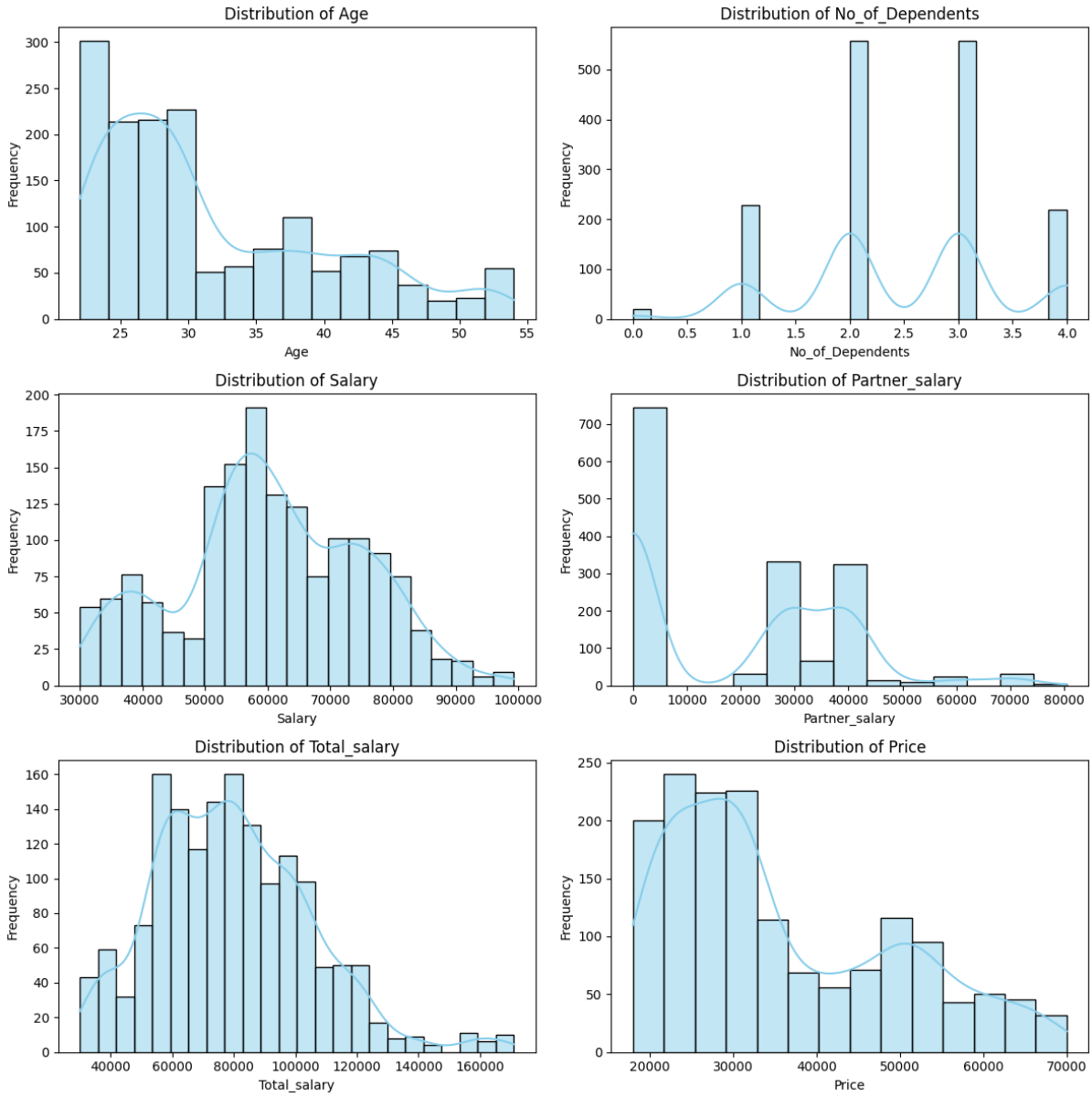

FIG 9: UNIVARIATE NUMERICAL ANALYSIS

Observation and Insights

- **Age:**

Most customers are aged 25–40, indicating the company targets young working adults.Distribution is slightly right-skewed, meaning fewer older customers.

- **Salary:**

  Salary ranges mostly between 50,000 and 80,000.A few high earners exist, creating a long tail on the right side.

- **Partner_salary:**

  A large number of values at 0, indicating many single-income families.Distribution becomes spread out after 20,000.

- **Total_salary:**

  Follows a right-skewed distribution with some high-income outliers.Most households earn 60,000–100,000.

- **Price:**

  Most customers are willing to spend 25,000–50,000.A small number are high spenders, possibly SUV buyers.

Exploring all the Categorical columns in data using Bar plot.

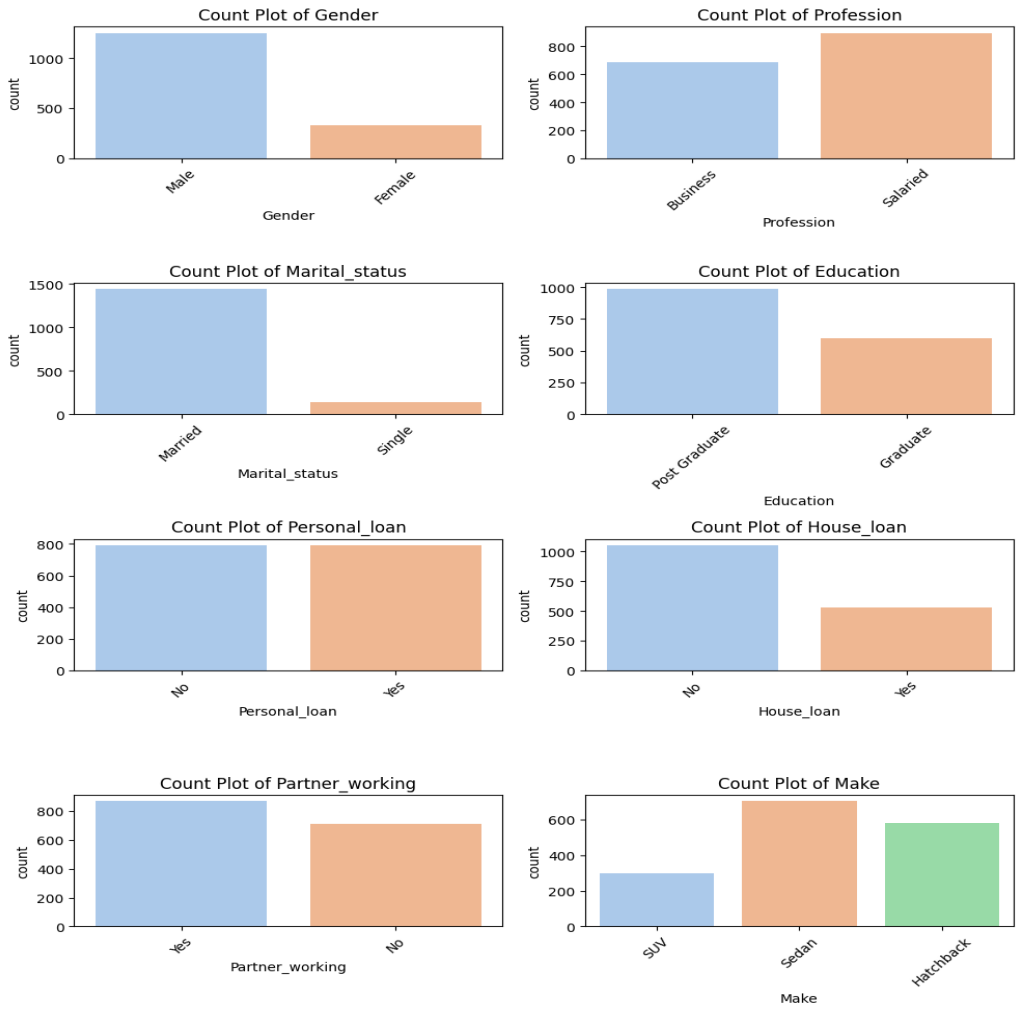

FIG 10: UNIVARIATE CATEGORICAL ANALYSIS

Observation and Insights

- **Gender:**

Males form the majority of the customer base.Marketing campaigns may target men more heavily.

- **Profession:**

The dataset includes customers from various professions; some may dominate (IT, Sales, etc.)Useful for segmentation.

- **Marital_status:**

Majority are married, which aligns with the presence of dependents.

- **Education:**

Mostly graduates and postgraduates, indicating a well-educated customer base.

- **Partner_working:**

A significant number of partners are not working, correlating with the 0 partner salary values.

- **Personal_loan / House_loan:**

Shows the financial responsibility levels of customers.Useful for determining purchase affordability.

- **Make:**

Shows customer preference among SUV, Sedan, Hatchback.Great for marketing insight.

5.2 CHECK FOR AND TREAT (IF NEEDED) OUTLIERS

Exploring all the numerical columns in data using Boxplot to find outliers

The boxplot of Total Salary shows a right-skewed distribution with high variability and several high-earning outliers. These outliers are meaningful and represent genuine premium customers. High-income families often prefer SUVs or higher-priced models, so removing them would remove valuable business insights. Since there is no indication that these values are incorrect, they are retained in the analysis.

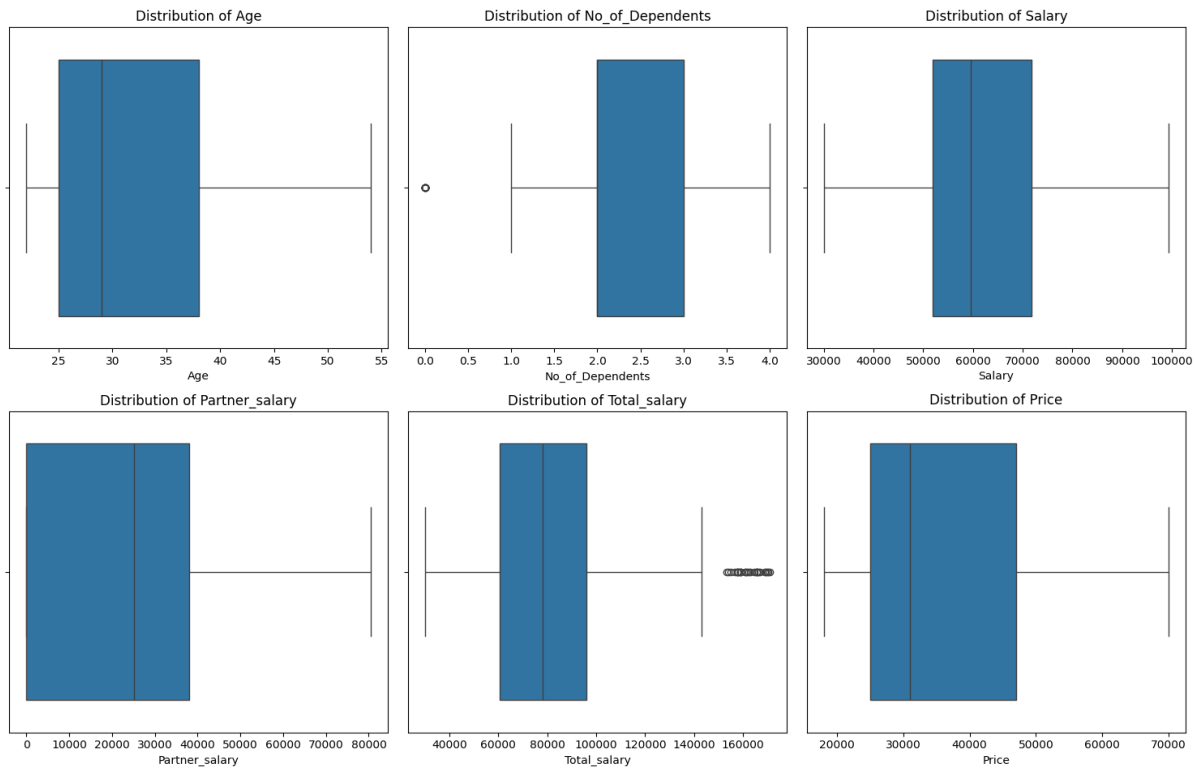

FIG 11: CHECKING OUTLIERS

6 BIVARIATE ANALYSIS

6.1 EXPLORE THE RELATIONSHIP BETWEEN ALL NUMERICAL VARIABLES

Bivariate analysis studies the relationship between two variables to identify meaningful patterns and associations. It helps reveal how factors like income, age, or loan status relate to car prices or customer choices. These findings support better understanding of customer behaviour.

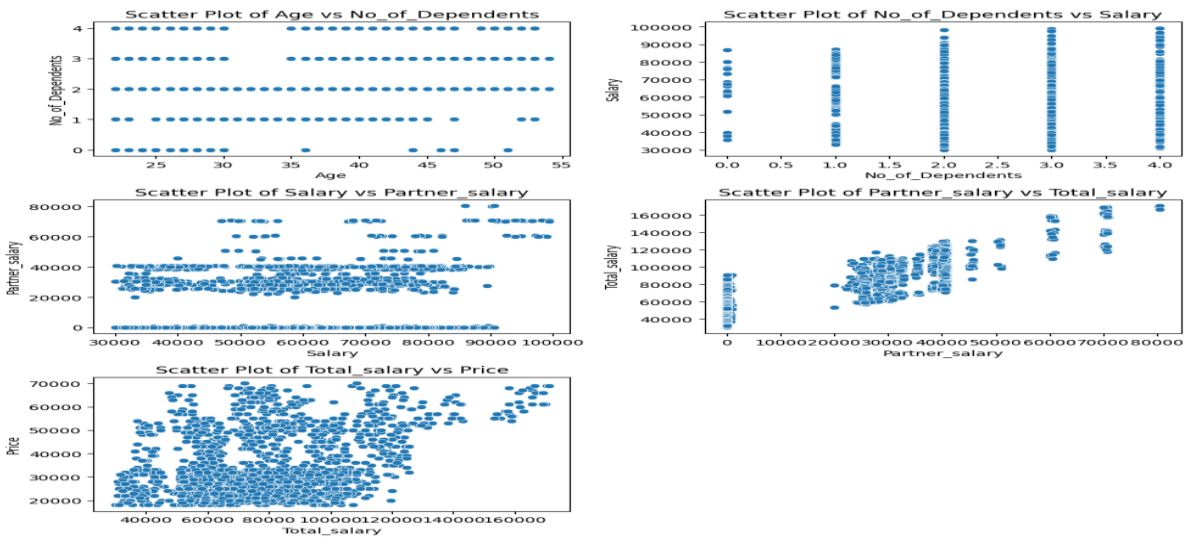

FIG 12:BIVARIATE NUMERICAL ANALYSIS 1

Although the initial bivariate analysis was performed on all numerical variables, further insights relevant to the marketing campaign required focusing on specific variable combinations. Therefore, we selected only the columns that directly influence customer purchasing behaviour, spending capacity, and car preference. This targeted approach ensures that the analysis remains aligned with the business objective and provides actionable insights for marketing decisions.

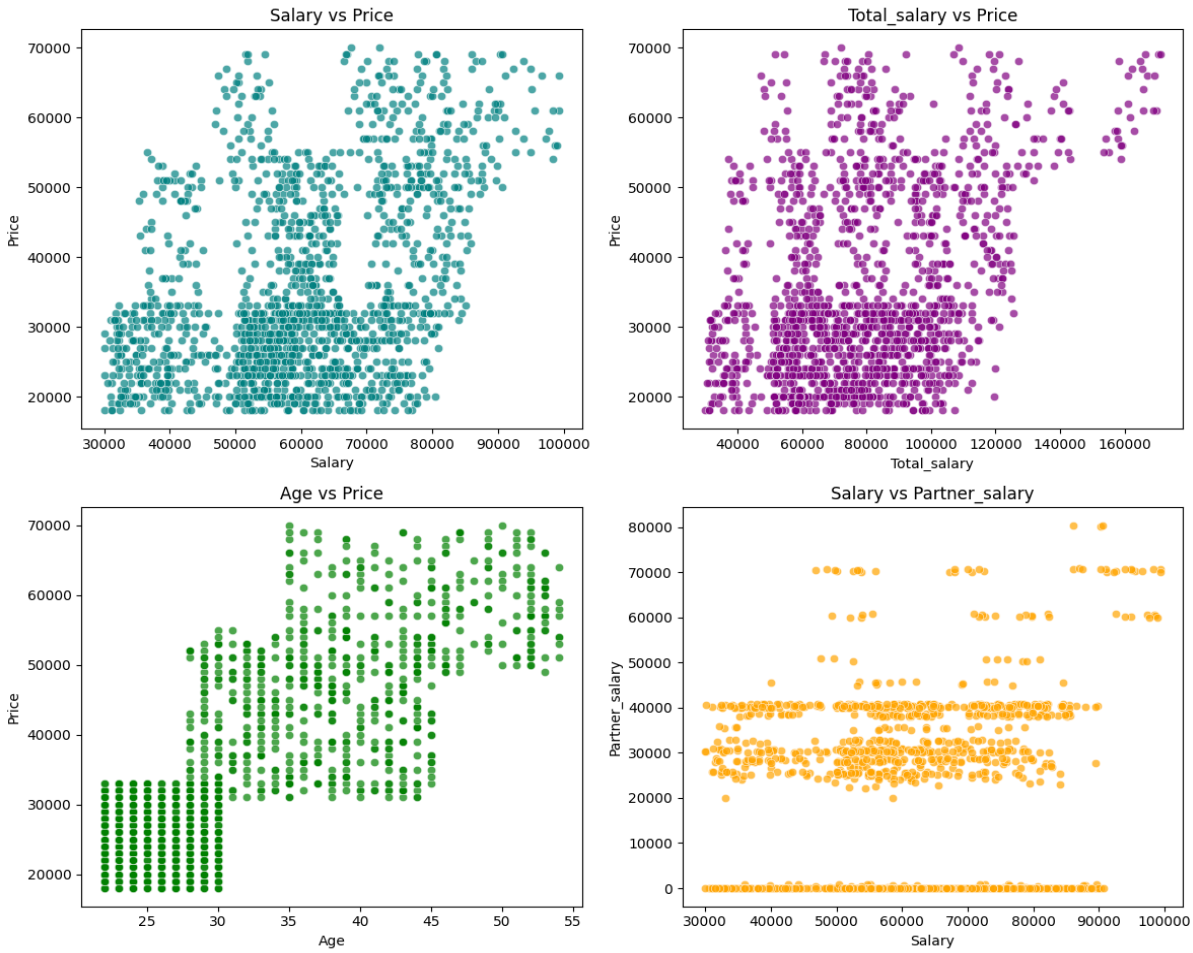

FIG 13:BIVARIATE NUMERICAL ANALYSIS 2

Observation and Insights

- **Salary vs Price:**

  The scatterplot shows a weak positive trend. Customers with higher salaries tend to buy slightly more expensive cars, but the points are widely spread. This means salary influences car price, but it is not the only factor. Other elements like personal preferences, loans, or family needs may also affect the purchase decision.

- **Total_salary vs Price:**

  This plot shows a clearer positive relationship compared to individual salary. As total household income increases, the price of the vehicle purchased also tends to increase. Higher-income families have a broader price range, meaning they can afford both mid-range and premium cars. This makes total household income a stronger predictor of purchasing power.

- **Age vs Price:**

  There is no strong visible pattern in this scatterplot. Customers of different ages (25–55) seem to purchase cars across similar price ranges. This indicates that age does not significantly influence the price of the car selected. Both younger and older individuals buy mid-range as well as higher-priced cars.

- **Salary vs Partner_salary:**

  The scatterplot shows mostly horizontal clustering, meaning partner salary remains low for many customers regardless of customer salary. This suggests that many households are single-income, or partner income does not scale with the customer's income. There is no strong relationship between the two variables.

6.2 EXPLORE THE CORRELATION BETWEEN ALL NUMERICAL VARIABLES

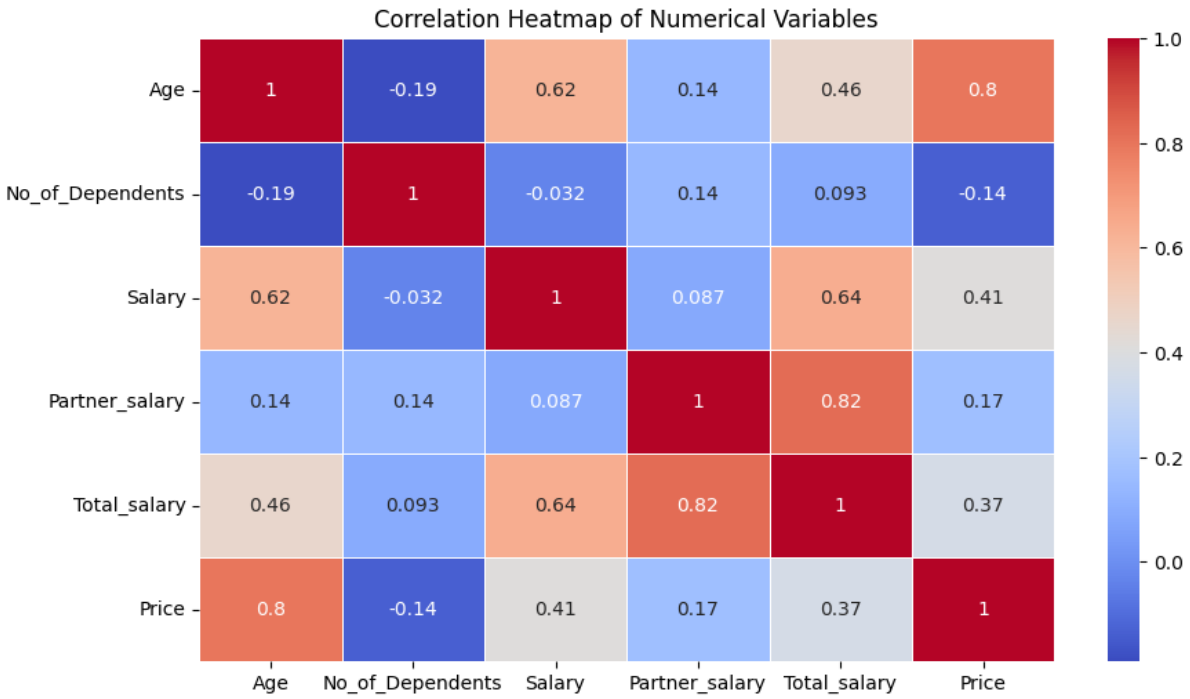

FIG 14 CORRELATION HEATMAP OF NUMERICAL VARIABLES

Observation and Insights

- **Salary and Total_salary (0.64-strong positive correlation):**

  Total salary increases as individual salary increases. This is expected because total salary is partly composed of the individual's salary. It confirms data consistency.

- **Partner_salary and Total_salary (0.82- very strong correlation):**

  Households with working partners have significantly higher total income. Partner salary is a major contributor to combined household earnings.

- **Age and Price (0.80-strong positive correlation):**

  Older customers tend to purchase higher-priced cars. This may be due to More financial stability,Higher savings,Preference for premium models. This is an important insight for the marketing team.

- **Salary and Price (0.41-moderate correlation):**

  As salary increases, customers are slightly more likely to choose higher-priced vehicles. However, the relationship is not very strong— indicating that car preference is influenced by factors beyond just income, such as:Brand loyalty, Family needs,Loan/EMI options.

- **Total_salary and Price (0.37-weak to moderate correlation):**

  Household income influences car price, but not strongly. This suggests customers across different income groups may still buy cars within similar price ranges.

- **No_of_Dependents (weak or negative correlations):**

  The number of dependents does not significantly affect Salary,Total income, Car price.Customers with dependents are not necessarily choosing cheaper or costlier cars.

- **Salary and Partner_salary (0.08- almost no correlation):**

  The partner's income is independent of the main customer's salary. This supports earlier findings that many households may be single-income.

6.3 EXPLORE THE RELATIONSHIP BETWEEN CATEGORICAL VS NUMERICAL VARIABLES

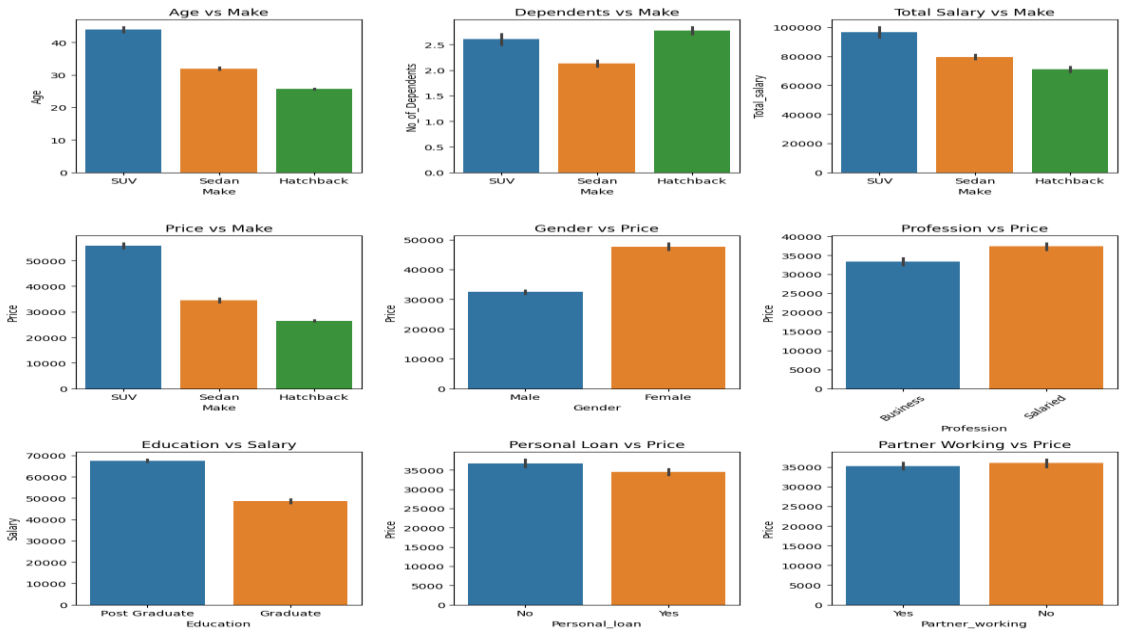

FIG 15: CATEGORICAL VS NUMERICAL ANALYSIS

Observation and Insights

- **Age vs Make:**

  SUV buyers have the highest average age, followed by Sedan buyers. Hatchback buyers are the youngest group.This means older customers prefer premium or larger cars (SUVs), while younger buyers choose Hatchbacks.

- **Dependents vs Make:**

  Hatchback buyers have the highest average number of dependents, followed by SUV and Sedan buyers.Families with more dependents are choosing compact or affordable cars, while SUVs attract medium-size families.

- **Total Salary vs Make:**

  SUV buyers show the highest total household income, followed by Sedan and then Hatchback buyers.Higher-income households prefer SUVs, while lower-income groups prefer Hatchbacks.

- **Price vs Make:**

  SUVs have the highest average price, Sedans mid-range, and Hatchbacks the lowest. Price distribution exactly matches car category market positioning.

- **Gender vs Price:**

  Females have a higher average car price compared to males. Female customers are spending more on cars, possibly choosing safer or better-featured models.

- **Profession vs Price:**

  Salaried professionals buy higher-priced cars compared to Business customers. Salaried employees show stronger purchasing power or preference for higher-end models.

- **Education vs Salary:**

  Postgraduates earn higher salaries compared to Graduates.Education level positively impacts earning potential.

- **Personal Loan vs Price:**

  Customers who have personal loans purchase slightly higher-priced cars than those without loans.Loans help customers buy better models than they otherwise could.

- **Partner Working vs Price:**

  Customers with a working partner have slightly higher car prices on average. Dual-income households have more financial flexibility, leading to higher spending.

7 KEY QUESTIONS

7.1 DO MEN TEND TO PREFER SUVS MORE COMPARED TO WOMEN?

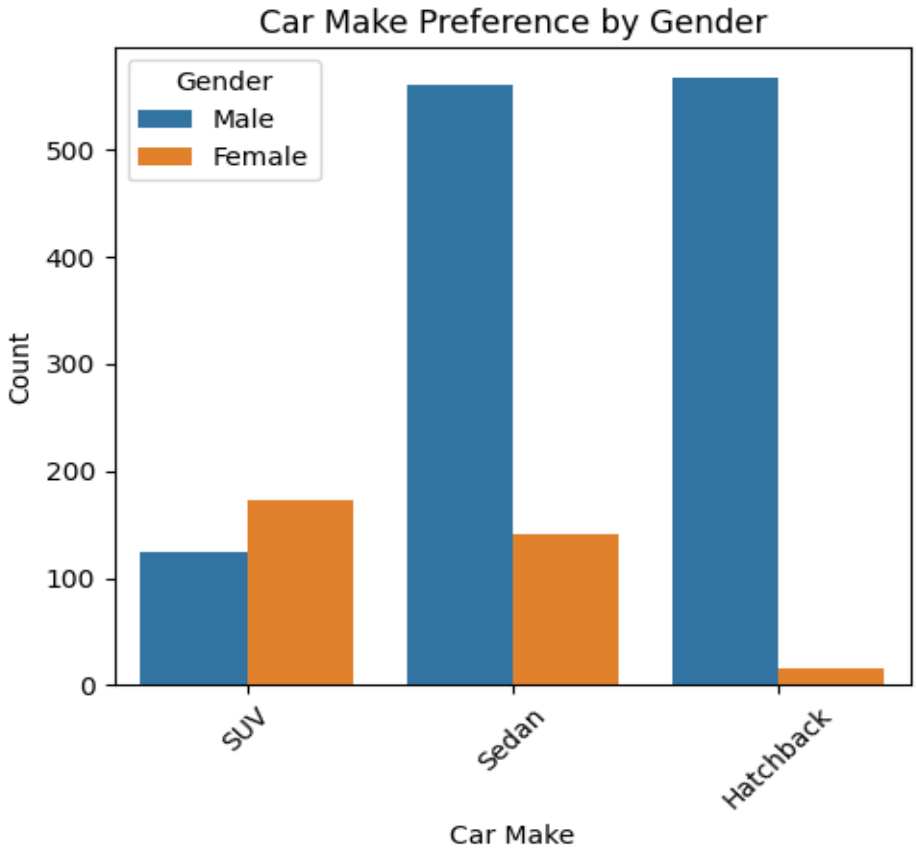

FIG 16:CAR MAKE PREFERENCE BY GENDER

For SUV, the female count (approx. 170) is higher than the male count (approx. 125).

For Sedan and Hatchback, male counts are much higher than female counts.

But the question is about SUV preference, and **females prefer SUVs more in this dataset**.

7.2 WHAT IS THE LIKELIHOOD OF A SALARIED PERSON BUYING A SEDAN?

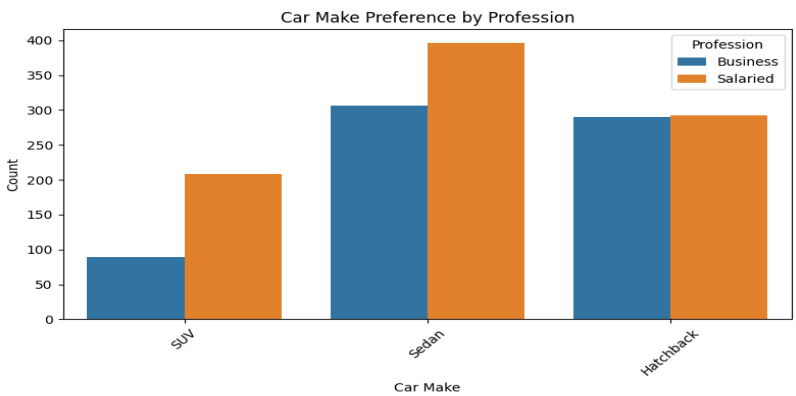

FIG 17:CAR MAKE PREFERENCE BY PROFESSION

The likelihood of **a salaried person buying a Sedan is high**, as the bar plot shows that salaried individuals have the highest count for Sedan compared to SUV and Hatchback.

7.3 WHAT EVIDENCE OR DATA SUPPORTS SHELDON COOPER'S CLAIM THAT A SALARIED MALE IS AN EASIER TARGET FOR A SUV SALE OVER A SEDAN SALE?

FIG 18:SUV OVER SEDAN BY SALARIED MALE

Sedan is the top choice for salaried males.

- Male Sedan count is the highest among all categories (approx. 300+). Males strongly prefer Sedans over SUVs or Hatchbacks.
- Females prefer SUVs more than other car types Female SUV count is the highest among female salaried customers. Female Hatchback count is extremely low.
- No evidence that a salaried male is an easier target for an SUV. For salaried males: Sedan >>Hatchback >> SUV (SUV is the lowest for males)

So, Sheldon Cooper's claim is not supported — salaried males do not prefer SUVs.

7.4 HOW DOES THE AMOUNT SPENT ON PURCHASING AUTOMOBILES VARY BY GENDER?

Gender	Price
Female	47705.167
Male	32416.134

dtype: float64

FIG 19:AMOUNT SPENT VARY BY GENDER

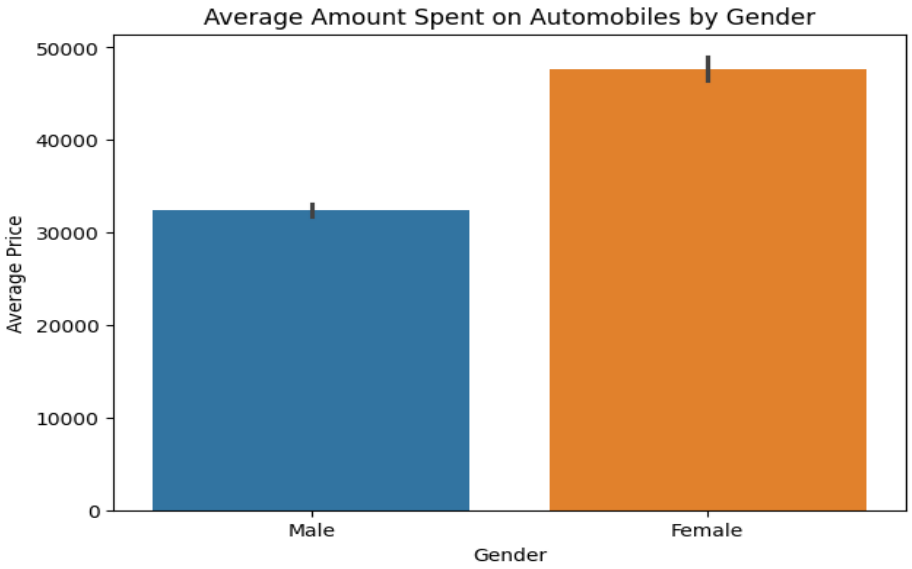

FIG 20: AVERAGE AMOUNT SPENT BY GENDER

Females spend more than males on automobile purchases.

Average amount spent by females approx. 47,705

Average amount spent by males is lower (as shown in the bar plot and groupby data)

7.5 HOW MUCH MONEY WAS SPENT ON PURCHASING AUTOMOBILES BY INDIVIDUALS WHO TOOK A PERSONAL LOAN?

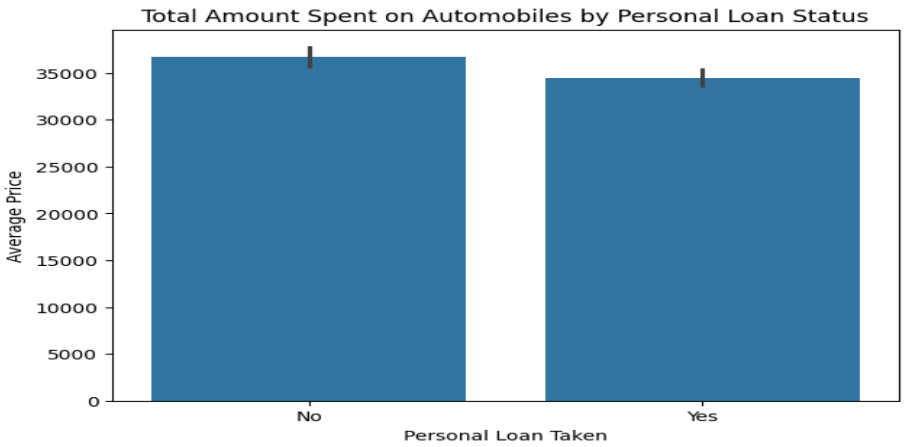

FIG 21: AMOUNT SPENT VARY BY PERSONAL LOAN STATUS

The bar plot compares the average car price based on whether a customer has taken a personal loan. Customers without a personal loan tend to purchase slightly higher-priced vehicles on average. Loan-takers generally spend a bit less, indicating that they may prefer more economical options due to financial constraints. **Overall, personal loan status appears to influence the price range of automobiles purchased.**

7.6 HOW DOES HAVING A WORKING PARTNER INFLUENCE THE PURCHASE OF HIGHER-PRICED CARS?

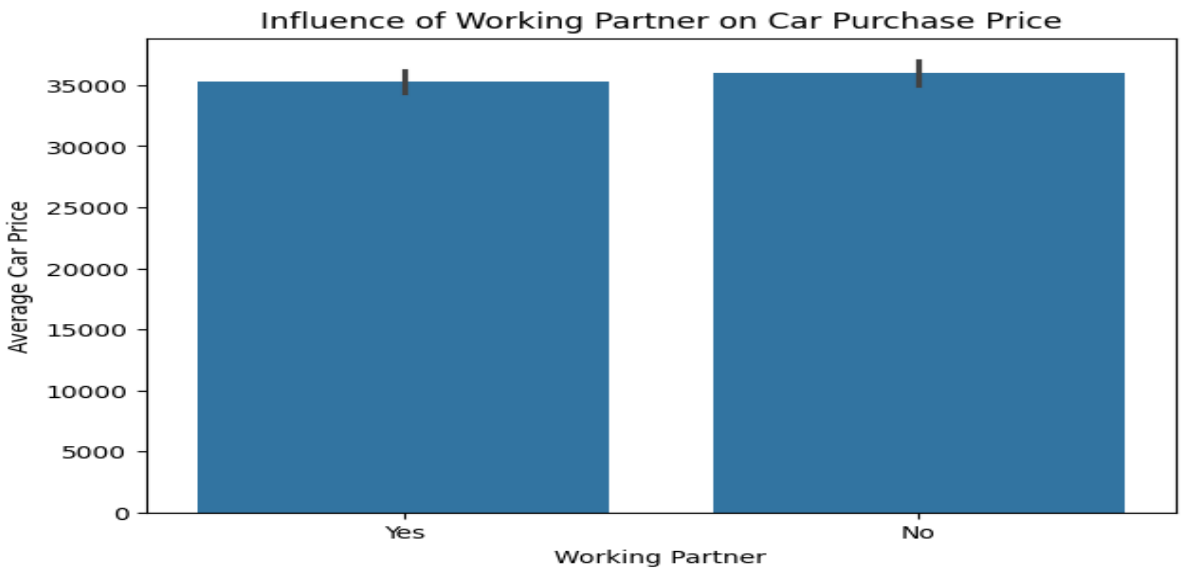

FIG 22: INFLUENCE OF WORKING PARTNER ON CAR PRICE

Based on the bar plot, having a working partner does not significantly influence the purchase of higher-priced cars. **The average amount spent by individuals with a working partner is almost the same as those without a working partner.**

8 ACTIONABLE INSIGHTS AND RECOMMENDATIONS

8.1 ACTIONABLE INSIGHTS

1. **Car preferences vary strongly by gender**

From the Make vs Gender analysis:

- Females prefer SUVs more than males, as shown by significantly higher female SUV counts.
- Males overwhelmingly prefer Sedans, with the highest male count in the Sedan category.
- Hatchbacks are purchased mostly by males, with minimal female involvement.

Implication: Gender plays an important role in car type preference.

2. **Profession significantly influences car choice From the Make vs Profession plot:**

- Salaried individuals strongly prefer Sedans, followed by Hatchbacks.
- Business professionals prefer Sedans and Hatchbacks almost equally, with very low SUV preference.

Implication: Sedans dominate across professions, but SUVs are less attractive to business professionals compared to salaried females.

3. **Salaried males are NOT easier targets for SUV sales From the multivariate breakdown:**

- Salaried males show the highest count for Sedans, not SUVs.
- Female salaried customers show more interest in SUVs.

Therefore, the assumption that a "salaried male is an easier target for SUV sales" is not supported by the data.

Implication: Marketing SUV campaigns toward salaried males would be ineffective.

4. **Female customers spend slightly more on average From the Average Price vs Gender plot:**

- Females spend an average of ₹47,705, which is higher than the male average.

This suggests that female buyers may opt for slightly higher-priced or feature-rich models.

Implication: Female customers represent a high-value segment for premium offerings.

5. **Individuals who took personal loans spent slightly less From the Personal Loan vs Price (sum) plot:**

- Customers without loans spent more in total than those who took loans.

This indicates loan-takers may be more price-sensitive or opt for budget-friendly models.

Implication: Loan-takers require more flexible finance schemes to encourage higher-value purchases.

6. **Working partner status has little to no impact on spending From the Working Partner vs Price plot:**

- Average car price is almost identical for customers with or without a working partner.

This indicates dual-income households do not significantly change car spending behavior in this dataset.

Implication: Partner's working status is not a strong predictor of high-value purchases.

7. **Hatchback purchases are driven mainly by males Across all segments:**

- Hatchback is dominated by male customers.
- Female hatchback purchases are extremely low.

Implication: Hatchbacks appeal to practical, possibly first-time male buyers

8.2 BUSINESS RECOMMENDATION

1. **Position SUVs in women-focused marketing campaigns, since female customers show stronger SUV preference:**

- Create women-centric promotional offers.
- Highlight features like safety, comfort, and family suitability.
- Use influencer marketing targeting female audiences.

Outcome: Higher SUV conversions among female buyers.

2. **Focus Sedan promotions on salaried and male customers, Since Sedans are most preferred by males and salaried professionals:**

- Provide special corporate EMI plans for salaried employees.
- Highlight performance, reliability, and comfort features.
- Offer loyalty discounts for professionally employed customers.

Outcome: Boost sedan sales in the strongest demand segment.

3. **Promote premium upgrades to female buyers, because females spend more on average:**

- Promote automatic variants, safety packages, and premium trims.
- Provide trial drives with premium models.
- Offer women-exclusive financing options.

Outcome: Higher revenue from premium product lines.

4. **Create targeted finance schemes for loan-takers, Loan-takers tend to spend less:**

- Introduce lower-interest auto loan packages.
- Provide bundled insurance + accessory discounts.
- Encourage "Upgrade Now, Pay Later" schemes.

Outcome: Increased spending and improved customer conversion.

5. **Avoid using working-partner status as a segmentation strategy, The dataset shows no meaningful difference in spending between:**

- Customers with a working partner
- Customers without one

Therefore:

- Do not allocate marketing budget to "dual-income" targeting.
- Focus instead on gender, profession, and loan status which are far stronger predictors.

6. **Strengthen marketing of Hatchbacks toward young male professionals, Since hatchbacks show male dominance:**

- Market a value-for-money positioning.
- Highlight mileage, maintenance cost, and compact design.
- Offer budget-friendly EMI plans for young customers.

Outcome: Improved penetration in the male entry-level segment.

7. **Use profession-based segmentation in advertising, Profession clearly influences car type choice:**

- Promote Sedans in corporate newsletters, LinkedIn ads, and IT park events.
- Promote SUVs and Hatchbacks through entrepreneurship communities and small business networks.

Outcome: More relevant targeting and higher engagement.